Congruence Rigidity and the Fusion Bound: what a positive weight can and cannot do to the spectrum of a transfer kernel

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Abstract

A positive site weight acts on a transfer kernel by *congruence*, $K \mapsto DKD$ with D positive diagonal, not by similarity. Similarity preserves the entire spectrum; congruence preserves strictly less, and strictly more than nothing. We determine both halves for the kernel of L decoupled Ising bonds. *Rigid half:* the sign of every quadratic form value survives, so definiteness in either sign is a congruence invariant — the definite case of Sylvester's law of inertia, recorded here in the elementary form that a machine can check without an eigenvalue count. *Fragile half:* the value of the spectral ratio $r = |\lambda_1|/|\lambda_0|$ does not survive at all, and we locate exactly how far it moves. Restricting the L-site kernel to the two antipodal configurations leaves *one* Ising bond of coupling βL; its ratio is $\tanh(\beta L)$. A weight concentrating on the ferromagnetic configurations therefore *fuses* the L sites into a single effective site carrying L times the coupling. Since $\tanh(\beta L) \to 1$, no bound on r can be uniform in L: the obstruction to volume-uniformity is named, exactly, and it is a property of the congruence and not of any spectral estimate. Consequently an argument that bounds r using only congruence invariants — rank, inertia, definiteness — cannot produce an L-uniform bound at all. We then ask which congruence is extremal. For *uniform* concentrations the answer is proved: keeping m sites of mutual correlation μ leaves $\mu J + (1-\mu)I$, of ratio $(1-\mu)/(1+(m-1)\mu)$, which is strictly decreasing in m — the more sites the weight keeps, the smaller the ratio it can reach, so the best uniform concentration is a *pair*. For arbitrary weights we record, with certified numerics and *without proof*, the conjecture that $\sup_{D>0} r(DMD) = (1-\mu)/(1+\mu)$ with $\mu = \min_{i\neq j} M_{ij}$: the extremal congruence concentrates on the *least correlated pair*. The hypercube instance is the fusion bound. All theorems are machine-checked in Lean 4.

1 The question

Let M be a real symmetric matrix and D an invertible diagonal matrix with positive entries. The map

$$M \longmapsto DMD$$

is a *congruence*. It arises whenever a statistical-mechanical transfer kernel is symmetrised against a site weight: if the unnormalised kernel is $w(\sigma)K(\sigma,\tau)$, the symmetric form with the same spectrum is $\sqrt{w(\sigma)}\,K(\sigma,\tau)\sqrt{w(\tau)}$, that is DKD with $D = \mathrm{diag}(\sqrt{w})$.

Congruence is not similarity. Similarity $M \mapsto SMS^{-1}$ preserves the characteristic polynomial and therefore everything spectral. Congruence preserves the quadratic form only up to a change of variable, and the change of variable is not an isometry. The natural question — and, we will argue, the one that governs whether a spectral estimate can be uniform in the system size — is:

Which spectral data survive a positive-diagonal congruence, and how far can the rest move?

We answer both parts for the kernel of L decoupled Ising bonds. Section 2 gives the rigid half, Section 3 the fragile half, Section 4 the consequence, and Section 5 states a conjectural general form with its numerical certificate.

What this paper does not do. It does not decide whether any particular coupled transfer operator has a spectral gap uniform in its extension. It *explains* why such a question is hard by exhibiting the exact mechanism that destroys uniformity, and explaining is not resolving. No statement here has a consequence for gauge theory in any dimension, and none is claimed.

2 The rigid half: congruence cannot move a sign

Write $q_M(x) = \sum_{i,j} x_i M_{ij} x_j$ for the quadratic form of M.

Lemma 2.1 (Change of variable)**.** *For every M, every d, and every x,*

$$q_{\operatorname{diag}(d)\, M\, \operatorname{diag}(d)}(x) \;=\; q_M(dx), \qquad (dx)_i := d_i x_i.$$

Proof. $(\operatorname{diag}(d) M \operatorname{diag}(d))_{ij} = d_i M_{ij} d_j$; substitute and reassociate. □

Theorem 2.2 (Definiteness is a congruence invariant)**.** *Let $d_i \neq 0$ for every i. Then* $\operatorname{diag}(d) M \operatorname{diag}(d)$ *is positive definite if and only if M is, and negative definite if and only if M is.*

Proof. By Lemma 2.1 the two forms have the same value set, because $x \mapsto dx$ is a bijection of $\mathbb{R}^n$ (its inverse is $x \mapsto d^{-1}x$) and carries nonzero vectors to nonzero vectors: if $d_i x_i = 0$ with $d_i \neq 0$ then $x_i = 0$. Apply this in both directions. □

Theorem 2.2 is the definite case of Sylvester's law of inertia [1], which says more: the full signature (n_+, n_0, n_-) is a congruence invariant. We do not reprove the general law. We isolate the definite case because it admits a proof through the form alone — no diagonalisation, no eigenvalue count — and that is what makes it cheap to certify mechanically.

Remark 2.3. Invertibility of D is not decorative. A diagonal with k zero entries drops the rank by k and can turn a definite form into a degenerate one. Every claim below assumes D strictly positive.

3 The fragile half: the fusion identity

Let $s(\sigma_j, \tau_j) = +1$ if $\sigma_j = \tau_j$ and -1 otherwise, and let

$$K_\beta^{(L)}(\sigma, \tau) \;=\; \exp\Big(\beta \sum_{j=1}^{L} s(\sigma_j, \tau_j)\Big), \qquad \sigma, \tau \in \{0,1\}^L,$$

the kernel of L decoupled Ising bonds at coupling β. Write $\sigma^+ = (0, \dots, 0)$ and $\sigma^- = (1, \dots, 1)$ for the two antipodal (ferromagnetic) configurations, and let

$$B(a) \;=\; \begin{pmatrix} e^a & e^{-a} \\ e^{-a} & e^a \end{pmatrix}$$

be the transfer matrix of a *single* Ising bond of coupling a.

Lemma 3.1 (One bond)**.** *$B(a)$ has eigenvectors $(1,1)$ and $(1,-1)$ with eigenvalues $2\cosh a$ and $2\sinh a$. Its spectral ratio is*

$$\frac{2\sinh a}{2\cosh a} \;=\; \tanh a.$$

Proof. Direct: $B(a)(1,1)^\top = (e^a + e^{-a})(1,1)^\top$ and $B(a)(1,-1)^\top = (e^a - e^{-a})(1,-1)^\top$. Both eigenvalues are real and $2\cosh a > 0$, so the ratio is well defined. □

Theorem 3.2 (Fusion identity)**.** *The 2×2 principal submatrix of $K_\beta^{(L)}$ on the antipodal pair $\{\sigma^+,\sigma^-\}$ is exactly one Ising bond of coupling βL:*

$$\begin{pmatrix} K_\beta^{(L)}(\sigma^+,\sigma^+) & K_\beta^{(L)}(\sigma^+,\sigma^-) \\ K_\beta^{(L)}(\sigma^-,\sigma^+) & K_\beta^{(L)}(\sigma^-,\sigma^-) \end{pmatrix} \;=\; B(\beta L),$$

and its spectral ratio is $\tanh(\beta L)$.

Proof. σ^+ agrees with itself at all L sites, so the sum of signs is $+L$ and the entry is $e^{\beta L}$; likewise for σ^-. The two antipodal configurations disagree at all L sites, so the sum is $-L$ and the entry is $e^{-\beta L}$. Compare with $B(\beta L)$ and apply Lemma 3.1. □

Theorem 3.2 is the mechanism this paper is about, and it is worth stating in words. A weight that concentrates on a set S of configurations does not *perturb* the kernel — it *restricts* it to S. When S is the antipodal pair, the restriction is a single bond whose coupling is L times the original. **The weight fuses the L sites into one effective site.**

4 Why that is a wall

Theorem 4.1 (No uniform ratio bound)**.** *For every $\beta > 0$ and every $\rho < 1$ there is an L with* $\tanh(\beta L) > \rho$. *Consequently the fused ratio is not bounded away from 1, and no estimate of the form $r \le \rho < 1$ can hold uniformly in the number of sites.*

Proof. $\tanh a = 1 - 2/(e^{2a}+1)$ is strictly increasing with limit 1; choose L with $e^{2\beta L} + 1 > 2/(1-\rho)$, which exists because $\beta > 0$ and $e^x > x$. □

This is the honest content of the paper's title. It is *not* a proof that any given coupled transfer operator fails to have a uniform gap; the concentration that realises the fusion is a limit, and the weight of a physical model may or may not approach it. What Theorem 4.1 establishes is that *the obstruction lives in the congruence*, not in whatever spectral estimate one happens to be attempting. An argument that bounds r using only data preserved by congruence — rank, inertia, definiteness — cannot possibly produce an L-uniform bound, because those data are identical for the unfused kernel (ratio $\tanh\beta$) and for every fused one (ratio up to $\tanh(\beta L)$).

Corollary 4.2. *Any L-uniform ratio bound must use a quantity that a positive-diagonal congruence can change.*

The orbit, not its closure

The fusion of Theorem 3.2 is realised by a weight that *concentrates*, and the congruences considered here are by *strictly positive invertible* diagonals. Setting the remaining diagonal entries to zero would leave the orbit and land in its closure, so the interface deserves to be stated rather than assumed. It is stated, and the statement is stronger than a limiting argument.

Proposition 4.3 (Concentration stays inside the orbit)**.** *Fix two configurations $p \neq q$ and $0 < \varepsilon \leq 1$, and let D_ε be the diagonal that is* 1 *at p and q and ε elsewhere. Then D_ε is strictly positive, and for* every *such ε,*

$$(D_\varepsilon M D_\varepsilon)_{pq} = M_{pq}, \qquad (D_\varepsilon M D_\varepsilon)_{qp} = M_{qp},$$

while $|(D_\varepsilon M D_\varepsilon)_{ij}| \leq \varepsilon\, |M_{ij}|$ whenever $i \notin \{p, q\}$.

Proof. $(D_\varepsilon M D_\varepsilon)_{ij} = d_i d_j M_{ij}$ with $d_p = d_q = 1$, giving the two equalities; and $d_i = \varepsilon$, $d_j \leq 1$ off the pair. □

So the fused bond is *already present at every strictly positive ε*: the limit does not create the block, it deletes the rest of the matrix. Converting that deletion into a statement about eigenvalues is the classical continuity of the spectrum in finite dimensions (eigenvalues depend continuously on the entries, e.g. [4, Appendix D]); we cite that step rather than reprove it, and Proposition 4.3 is where the citation attaches.

5 Which congruence is extremal: pairs beat exchangeable clusters

Theorem 3.2 exhibits *one* congruence reaching $\tanh(\beta L)$. Is it the best one? For the family of *uniform* concentrations the question has a complete and elementary answer, and the answer explains the mechanism.

Theorem 5.1 (Exchangeable concentration)**.** *Fix $0 < \mu < 1$ and let $E_m = \mu J_m + (1 - \mu) I_m$ be the exchangeable correlation matrix on m sites. Its spectrum consists of exactly two points: $1 + (m - 1)\mu$ on the constant vector, and $1 - \mu$ on every vector summing to zero. Hence the ratio it reaches is*

$$\frac{1 - \mu}{1 + (m - 1)\mu},$$

which is strictly decreasing *in m.*

Proof. $E_m \mathbf{1} = (1 + (m - 1)\mu)\mathbf{1}$ by summing a row. If $\sum_i x_i = 0$ then $(E_m x)_i = \mu \sum_j x_j + (1 - \mu)x_i = (1 - \mu)x_i$. These two eigenspaces have dimensions 1 and $m - 1$, so they exhaust the spectrum. For monotonicity, $k \mapsto (1 - \mu)/(1 + k\mu)$ is strictly decreasing on $k \geq 0$ because the numerator is positive and the denominator strictly increasing. □

The reading is the point. **The more sites an exchangeable concentration keeps, the smaller the ratio it can reach.** So among these the best is the smallest nondegenerate one — a *pair* — and at $m = 2$ the value is $(1-\mu)/(1+\mu)$. On the hypercube kernel, $\mu = e^{-2\beta L}$ at the antipodal pair, and $(1 - \mu)/(1 + \mu) = \tanh(\beta L)$: Theorem 3.2 is the $m = 2$ case of Theorem 5.1.

Remark 5.2 (Exactly how far this goes)**.** Theorem 5.1 concerns subsets on which *all* pairwise correlations equal a common μ. On a general subset the entries M_{ij} differ and the spectrum has no such closed form, so the theorem does *not* say that a pair beats every uniformly weighted subset of an arbitrary M. The section title should be read with the word *exchangeable* attached. What the theorem does establish is the mechanism — spreading a concentration over more equally correlated sites lowers the reachable ratio — and it is that mechanism, not a general proof, that makes Conjecture 5.3 plausible.

That settles the exchangeable family. For arbitrary weights we have only a conjecture, and we say so wherever it appears.

Conjecture 5.3 (Least-correlated pair). *Let M be symmetric with $M_{ii} = 1$ and $0 < M_{ij} < 1$ for $i \neq j$, and let $\mu = \min_{i \neq j} M_{ij}$. Then*

$$\sup_{D>0 \text{ diagonal}} r(DMD) \;=\; \frac{1-\mu}{1+\mu},$$

approached by concentrating D on an argmin pair, and not attained.

The lower bound is elementary: concentrating on a pair $\{i, j\}$ leaves $\left(\begin{smallmatrix} 1 & M_{ij} \\ M_{ij} & 1 \end{smallmatrix}\right)$ up to scale, of ratio $(1 - M_{ij})/(1 + M_{ij})$, which is largest at the least correlated pair. Theorem 5.1 adds that no *uniform* concentration on more sites can beat it. **The upper bound over arbitrary weights is open.** We have no proof, and we say so wherever the statement appears.

For the record, the reduction we did obtain and could not close: since T is positive definite exactly when M is (Theorem 2.2),

$$r(T) \le \frac{1-\mu}{1+\mu} \iff \lambda_0 - \lambda_1 \;\ge\; \mu\,(\lambda_0 + \lambda_1),$$

and writing $M = \mu J + (1 - \mu)N$ — where N again has unit diagonal and entries in $[0, 1)$ — gives $T = \mu\, dd^\top + (1 - \mu)\, DND$, a rank-one positive semidefinite piece plus an entrywise nonnegative one. Weyl's inequality and the Perron root of the second piece bound $|\lambda_1|$ correctly but lose too much on λ_0: testing against the Perron vector of DND reduces the claim to $\sum_{i \neq j}(1 + \mu - 2M_{ij})u_i u_j \ge (1 - \mu) \sum_i u_i^2$, whose left-hand side is negative once some $M_{ij} > (1 + \mu)/2$. The top eigenvector of T is not the Perron vector of DND, and the slack is exactly that gap.

On the hypercube, $K_\beta^{(L)}$ normalised to unit diagonal has $M_{\sigma\tau} = e^{-2\beta h(\sigma,\tau)}$ with h the Hamming distance; the least correlated pair is the antipodal one, $\mu = e^{-2\beta L}$, and $(1-\mu)/(1+\mu) = \tanh(\beta L)$. Conjecture 5.3 therefore contains Theorem 3.2 as its hypercube instance.

Numerical certificate. Random search plus Nelder–Mead ascent over positive diagonals, on families chosen to break the conjecture, never exceeded $(1 - \mu)/(1 + \mu)$ and in almost every case attained it to machine precision.

family	worst excess over target	cases
hypercube, $L \in \{2, 3, 4\}$, $\beta \in \{0.15, 0.4, 0.8\}$	1.0×10^{-15}	9
random positive Gram, $n \in \{4, 5, 6\}$	5.0×10^{-16}	12
full rank / near singular ($\kappa \sim 6 \times 10^6$)	3.3×10^{-16}	4
near-exchangeable, isolated argmin, entries near 1	4.4×10^{-16}	3
exactly exchangeable $\mu J + (1 - \mu)I$	1.0×10^{-15}	2
indefinite, $\lambda_{\min} \in [-0.42, -0.06]$	1.6×10^{-15}	12
larger sizes, $n \in \{8, 10, 12\}$, κ up to 5×10^4	4.2×10^{-16}	6

The last row was added after an audit of our own procedure: the adversarial sweep as first written never actually reached $n = 8$. Its positive-definiteness *precondition* fired on that family, the generator was repaired, and the family was not re-run — so the largest size with evidence had been $n = 6$, not $n = 12$ as the sweep's summary implied. A precondition that is never met carries no evidence either way, and the script now returns a distinct INCONCLUSIVE code rather than reporting such a case as a result.

The last row deserves comment, because it corrected us. A pre-registered gate predicted that positive definiteness would be load-bearing — that at least one of twelve indefinite witnesses would break the bound. None did; all twelve landed *on* it. The hypothesis was therefore dropped, and Conjecture 5.3 as stated above assumes no definiteness. The gate, its prediction, and its failure are recorded in the repository rather than repaired in place.

Scope of the certificate. These are numerics, not proof. They cover $n \leq 12$, $\mu > 0$, and $\lambda_{\min} \geq -0.43$. Nothing here tests $\mu \leq 0$ (where the target exceeds 1 and the question changes character), nor a negative eigenvalue large enough to make $|\lambda_0|$ the negative extreme, at which point r measures something else.

6 Relation to prior work

Sylvester (1852). The law of inertia is classical and Theorem 2.2 is its definite case. We claim no novelty on the rigid half; the contribution is that this form of it is checkable without diagonalisation.

Ostrowski. Ostrowski's quantitative form of the inertia law gives, for Hermitian A and invertible S, $\lambda_k(SAS^*) = \theta_k \lambda_k(A)$ with $\theta_k \in [\lambda_{\min}(SS^*), \lambda_{\max}(SS^*)]$. For $S = D$ diagonal this yields

$$r(DAD) \; \leq \; \kappa(D^2)\, r(A), \qquad \kappa(D^2) = \frac{\max_i d_i^2}{\min_i d_i^2},$$

which **diverges** exactly in the regime of interest, namely when D degenerates. Theorem 3.2 and Conjecture 5.3 supply a *finite* substitute in that regime: the supremum is $\tanh(\beta L)$, respectively $(1-\mu)/(1+\mu)$, however badly conditioned D becomes. That contrast — divergent classical bound, finite exact value — is the delta of Section 5, and it is stated here side by side so that it can be checked.

Onsager and Kaufman. The kernel $DK_\beta D$ with a ferromagnetic nearest-neighbour weight is the transfer matrix of the two-dimensional anisotropic Ising model on a strip, whose spectrum is known in closed form. Nothing in Sections 3–4 is new mathematics about that model, and the reader who knows the exact solution will find $\tanh(\beta L)$ unsurprising: it is the zero-temperature limit in the transverse direction, where the strip collapses to a single spin. The contribution is not the value. It is that the value is the supremum *within the family of uniform concentrations* (Theorem 5.1) and a lower bound for the full diagonal orbit, that the orbit's rigid invariants are simultaneously pinned, and that the pair of facts identifies which quantities an argument may and may not use. We are careful with that phrasing: whether $\tanh(\beta L)$ is the supremum over *all* positive diagonals is Conjecture 5.3 and is not proved here.

Formalisation. We are not aware of a mechanised treatment of congruence orbits of transfer kernels. This is an absence of found prior art in the searches we ran, not a claim of priority.

7 What is open

1. The upper bound in Conjecture 5.3 *over arbitrary weights.* Theorem 5.1 settles it for uniform concentrations; the gap is everything else. A proof would make the fusion bound an exact classification rather than an attained lower bound with a certificate, and §5 records where our attempt loses.

2. Whether the physical weight w_γ attains the antipodal restriction in the limit $\gamma \to \infty$ with the rate we measure. The deficit $\tanh(\beta L) - r(\gamma)$ was found to decay like $e^{-2\gamma}$ (fitted exponents -1.96 to -2.08 against a pre-registered prediction of -2), consistent with one-domain-wall configurations dominating the correction. The limit itself is not proved here.

3. The general signature-counting law as a mechanised statement.

Machine verification — status

Every theorem, lemma and proposition stated above — the rigid half (Theorem 2.2, Lemma 2.1), the one-bond spectrum (Lemma 3.1), the fusion identity (Theorem 3.2), the wall (Theorem 4.1), the orbit statement (Proposition 4.3) and the exchangeable computation (Theorem 5.1) — is written in Lean 4 against a pinned Mathlib in `YangMills/OS/CongruenceSpectrum.lean`, and **has been verified**. Two things are not, and both are labelled: the continuity of eigenvalues invoked after Proposition 4.3, which is classical and cited; and Conjecture 5.3, which carries the word *conjecture* in every sentence that states it.

commit	`d4126d34`
module `sha256`	`4a6c4d680becd37acc9d02b6` `5e0995ff0dc26c5cd7f4edfa51468a6c04029793`
toolchain	`leanprover/lean4:v4.29.0-rc6`
Mathlib pin	`07642720480157414db592fa85b626dafb71355b`
elaboration	exit `0`; empty log — zero errors, zero warnings, zero `sorry`
axiom oracle	29/29 declarations on `[propext, Classical.choice, Quot.sound]`
`sorryAx`	none

The module hash was computed independently on the author's machine and on the verification host and agrees, so the checked object is the source printed here. Compilation ran on a CPU (never GPU) high-memory cloud runtime, per the project's environment rule of 1 August 2026; the desktop compiles nothing.

One registered check remains undone. The pre-registration also asked that the module be imported into the project's aggregate library and that the build job count increment by exactly one. It has not been imported, so there is no increment to measure; measuring it honestly requires two full builds of the aggregate in the same tree, which the shared Mathlib cache does not cover. That check is therefore reported as *not run* rather than dropped, and it is not replaced by a comparison against a count measured elsewhere. Nothing in the theorems above depends on it.

References

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